SMART FARM MONITORING · GUIDE FOR FARM USERS

Your farm watches itself

Sensors check the soil, the air and the feed day and night.
Pumps, fans and feeders start by themselves, and you get a message when something needs you.

For farm owners, workers and technicians

WHY

Checking a farm by hand finds problems too late

1

The problem

Dry soil, a hot chicken house or an empty feeder are often found hours later: plants and animals suffer, water and feed are wasted.

2

What Smart Farm does

Small sensors in each bed, greenhouse and chicken house measure all the time. When something is wrong, the right machine starts.

3

What you get

A reaction in seconds instead of hours, fewer losses, less walking around, and a record of everything that happened.

HOW IT WORKS

Measure, decide, act, tell you

1

Measure

Sensors send soil moisture, temperature, humidity, light, feed and water levels every few seconds.

2

Decide

Your rules say what is normal, for example “soil wetter than 30 %”.

3

Act

Outside the normal range, the pump, fan, heater, mister, light or feeder starts, and stops when it is fixed.

4

Tell you

An alert appears in the app and, if you like, on Telegram on your phone.

WHO USES IT

Everyone sees what they need, and only that

Person	What they do in the app
Farm owner	Everything: zones, rules, schedules and staff accounts
Farm worker	Watches the farm, starts and stops machines, marks alerts as seen
Technician	Installs devices, reports broken ones and repairs them
Company staff	Set up new customer farms and help them, only when the owner allows it

Each person signs in with their own password. A worker never sees a button they are not allowed to use.

EVERY DAY

The whole farm on one screen

A greeting, the average temperature and a short summary of how the farm is doing.

Each zone shows its values; the most stressed zones come first.

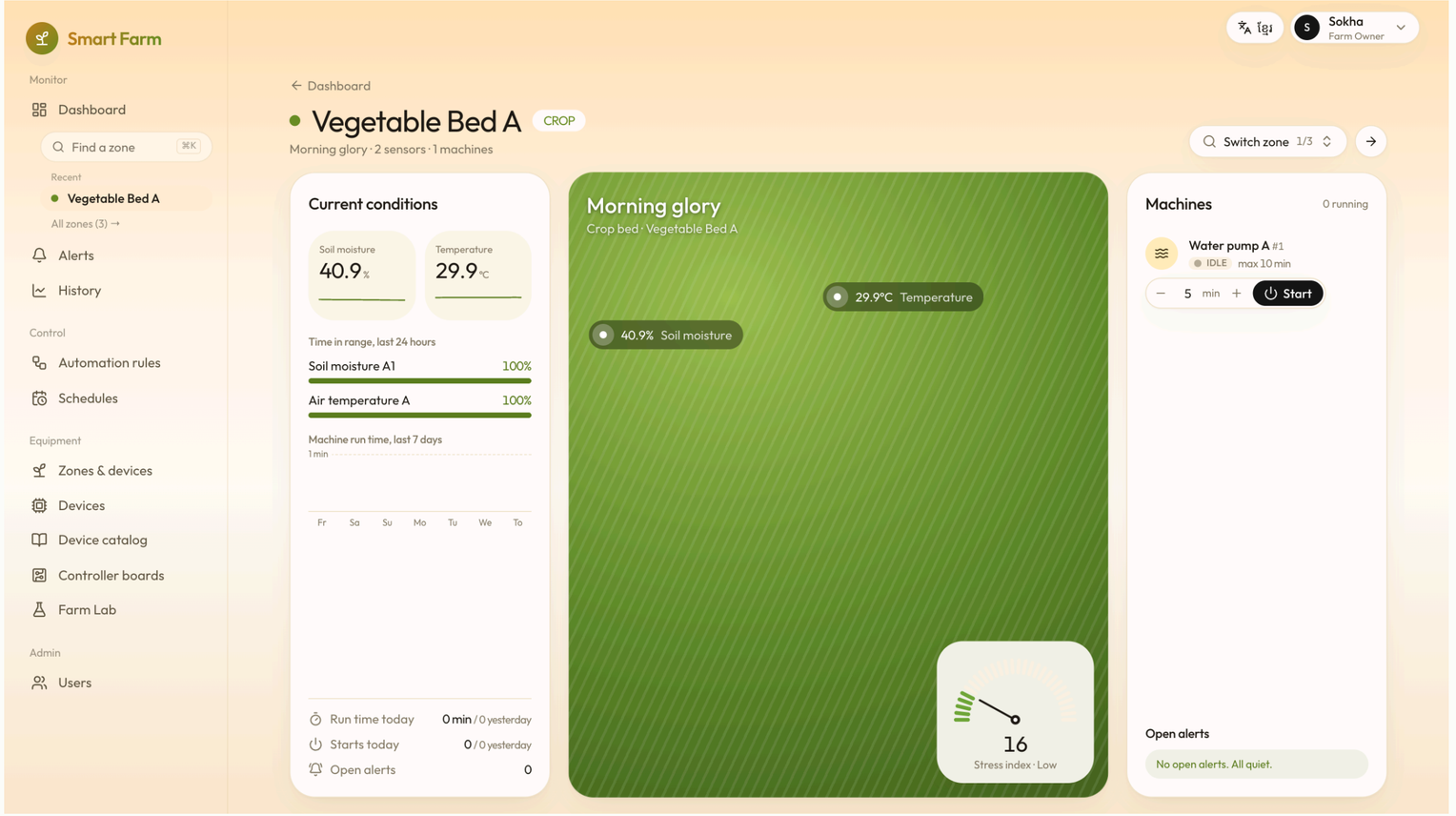


EVERY DAY

Each zone has its own page

The current values, how long today it stayed in the good range, its machines and alerts.

Below: a chart of the last 24 hours, how long each machine ran, and what happened.

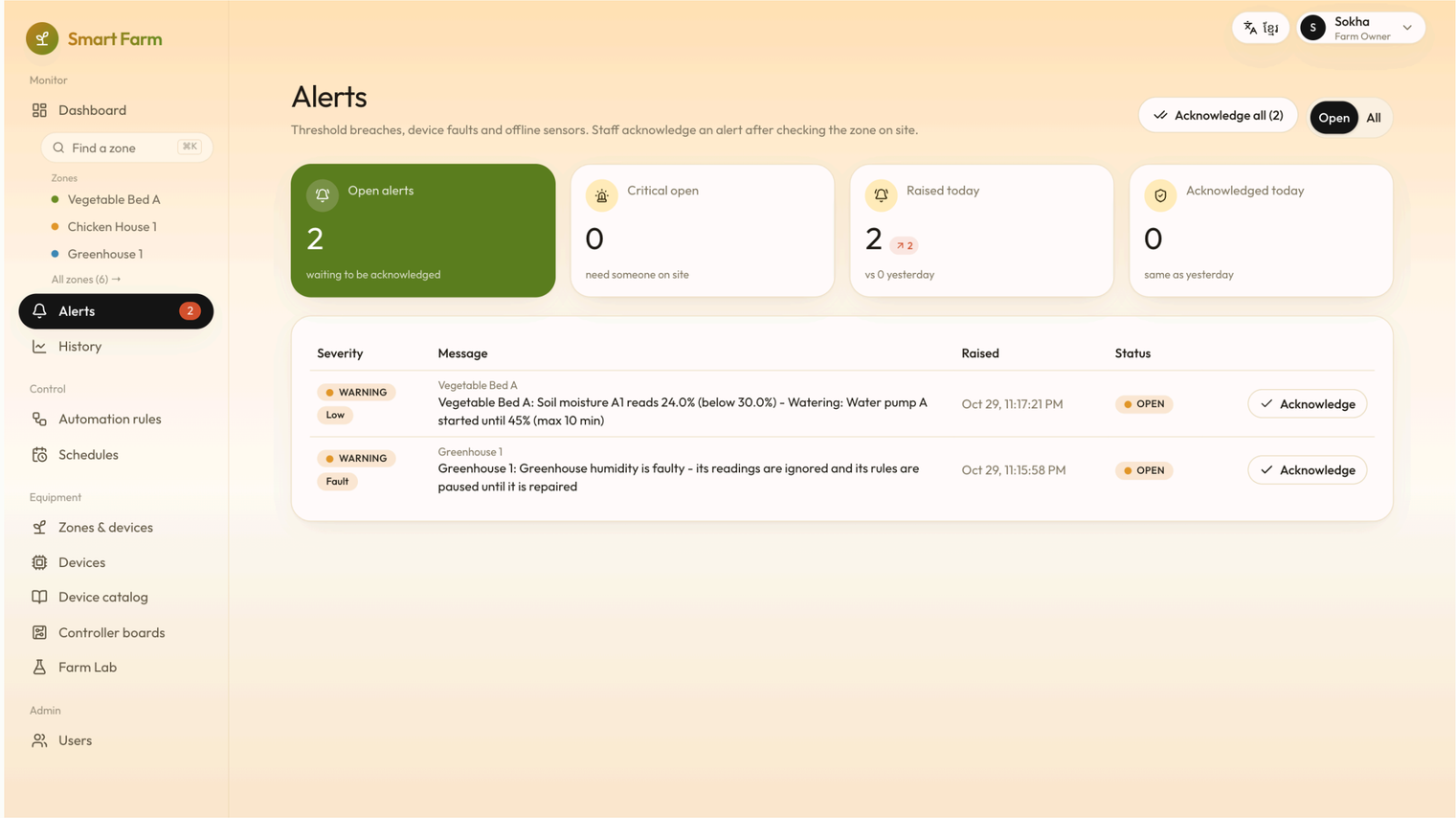


WHEN SOMETHING IS WRONG

Alerts tell you what happened and what was done

“Vegetable Bed A: Soil moisture A1 reads 24.0% (below 30.0%) – Watering: Water pump A started”

Mark an alert as seen, and the app shows when the value is back in range.



RULES

It keeps the chicken house comfortable, like a thermostat

Too hot: above 32 °C the fan starts, and it keeps running until the house is back to 29 °C.

Too cold: below 18 °C the heater starts, and stops at 21 °C.

Never both: the fan and the heater never run at the same time. Each machine also has a longest run, as a safety limit.

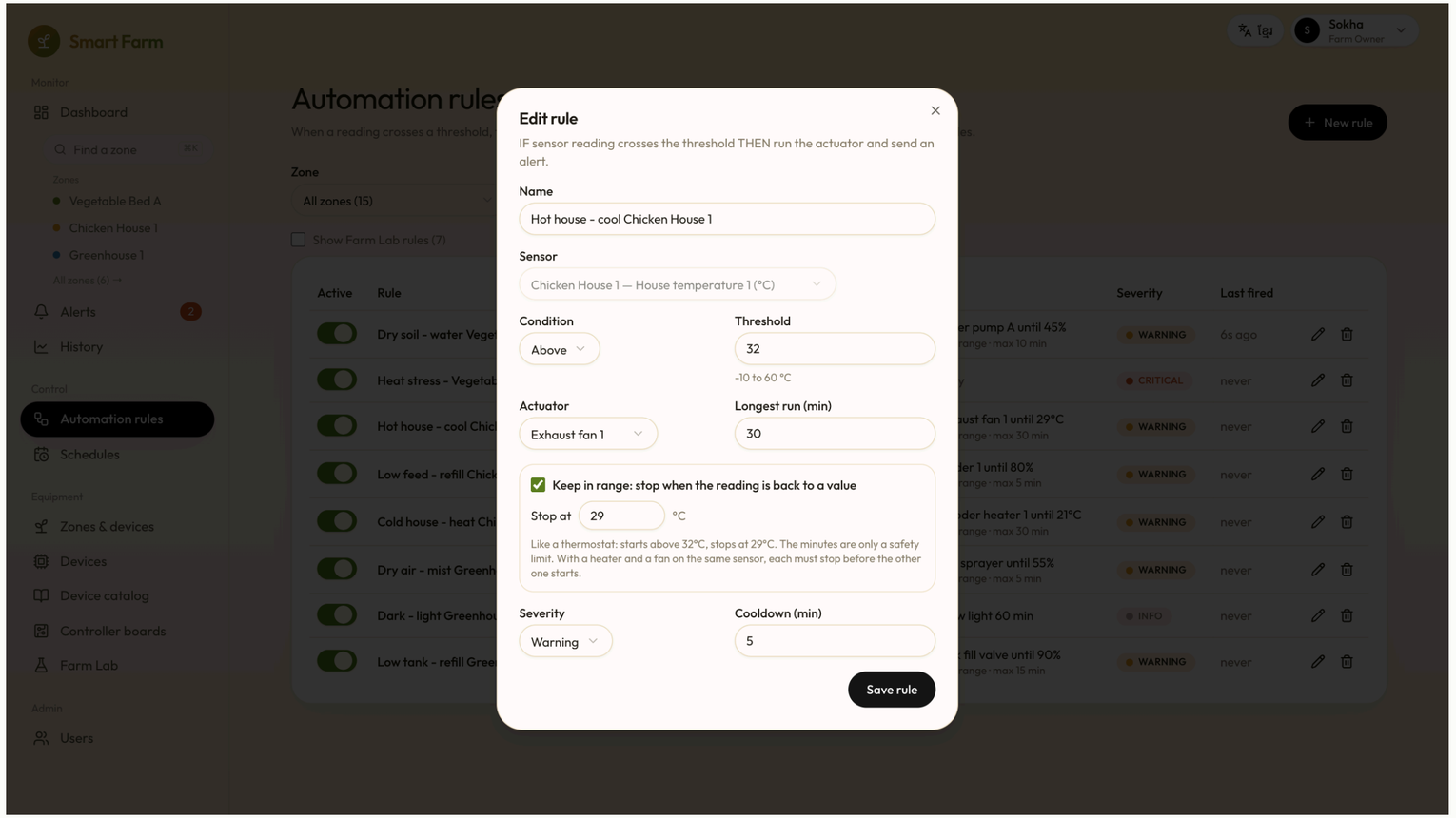
above 32 °C	fan starts
29 °C	fan stops
21 – 29 °C	comfortable
21 °C	heater stops
below 18 °C	heater starts

RULES

Setting a rule is one short form

Choose the sensor, when to start, the machine, and when to stop.

The form explains the rule in a sentence before you save it.

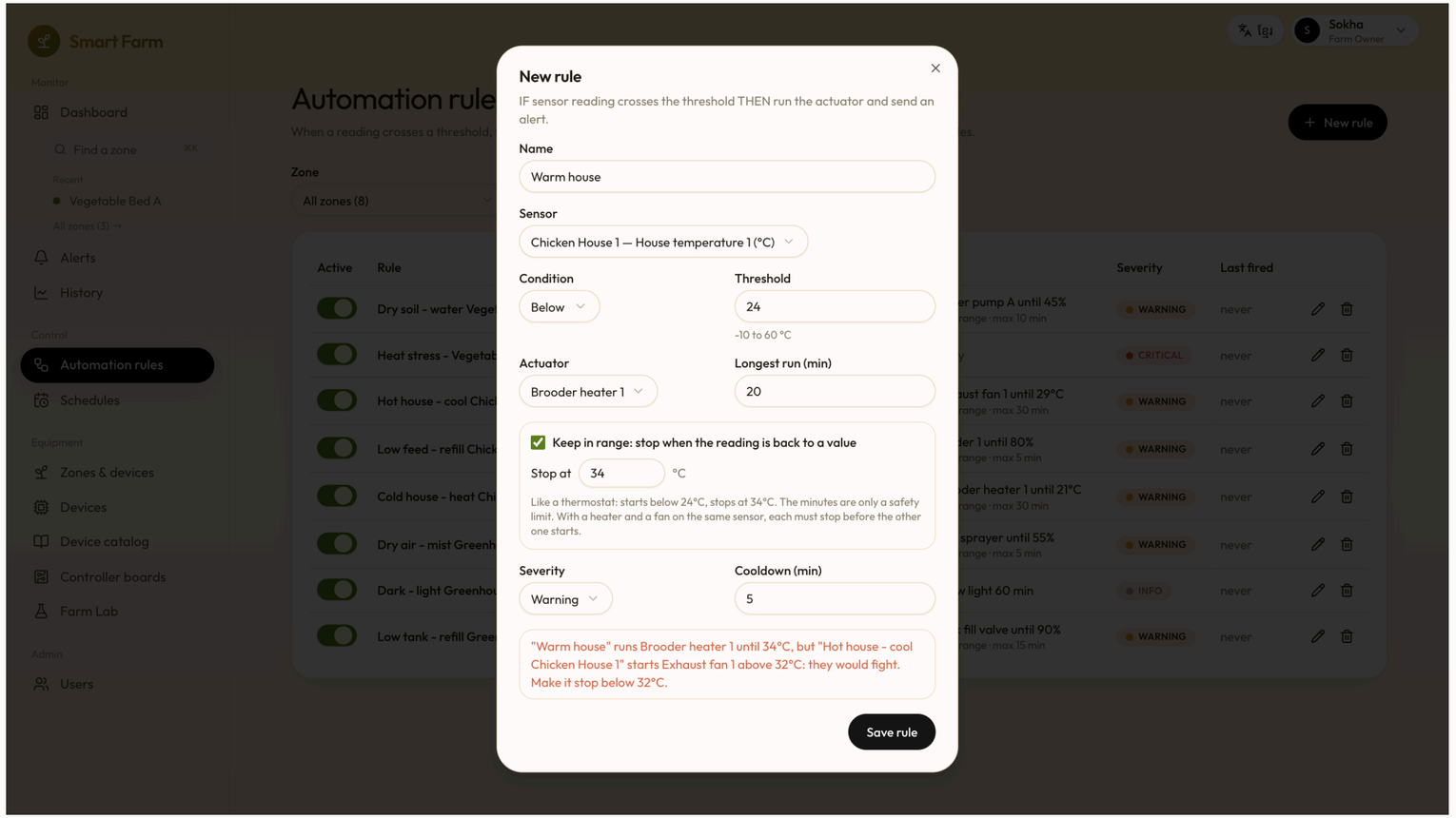


RULES

Rules that would fight are refused

A heater that runs up to 34 °C would fight the fan that starts at 32 °C.

The app says which two rules clash and what to change.

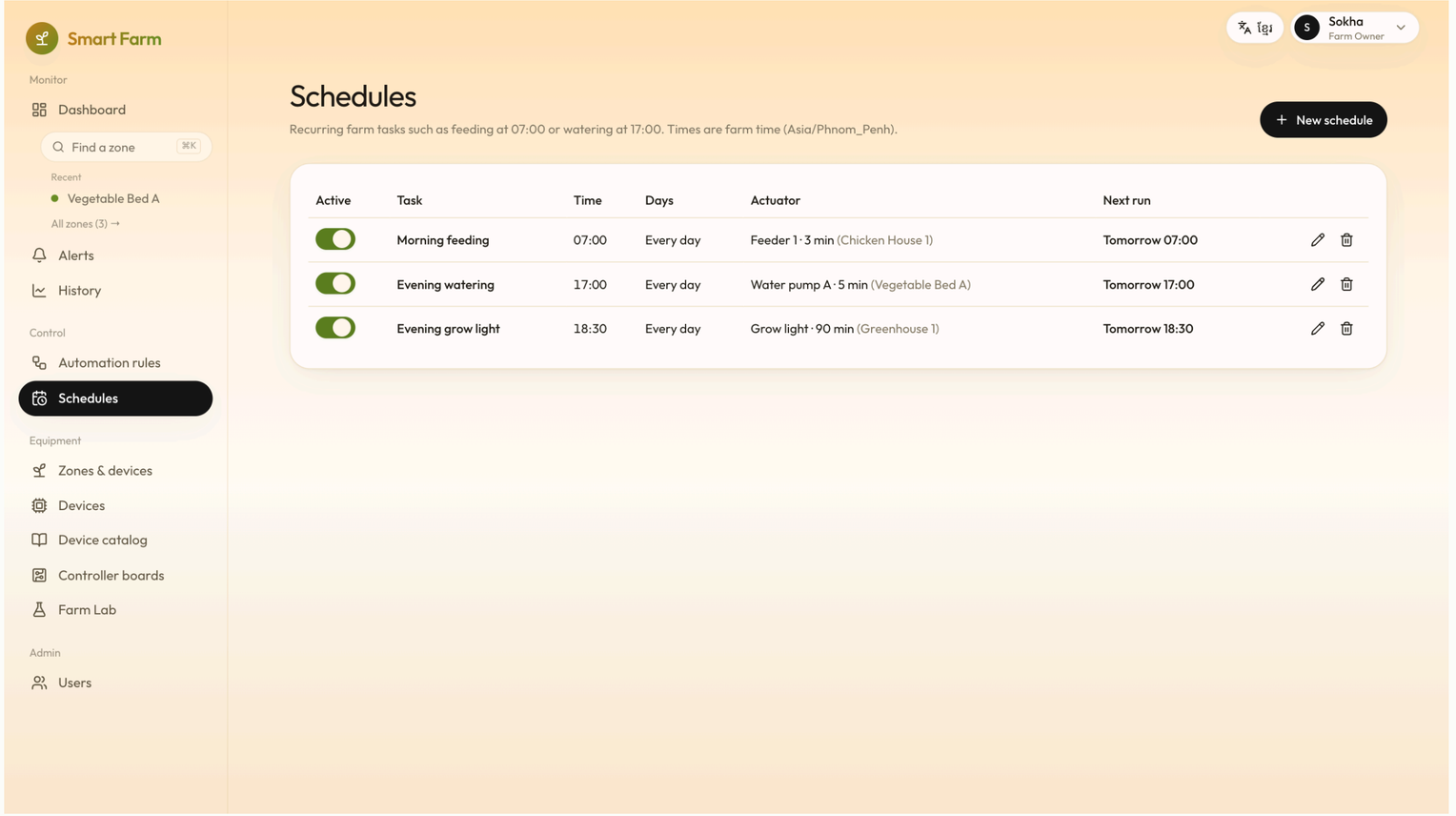


ROUTINE WORK

Schedules for the jobs you do every day

Feed the chickens every morning at 7:00, water the beds at 17:00.

Schedules follow the farm's own clock, Phnom Penh time.

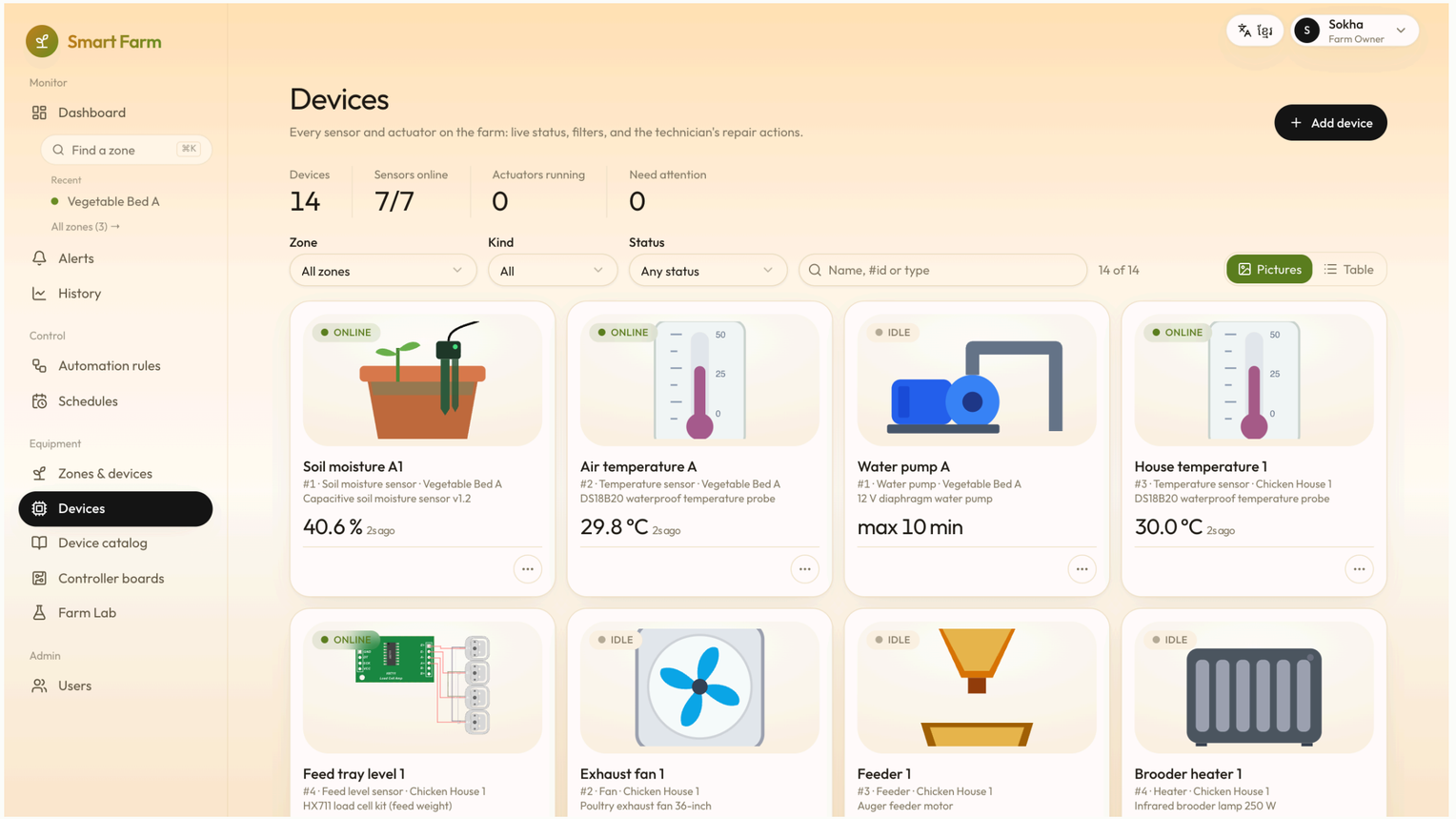


EQUIPMENT

Every device on one page

Each sensor and machine with its status and latest value.

“Need attention” shows only what is broken or offline.



EQUIPMENT

A broken sensor cannot fool the farm

1

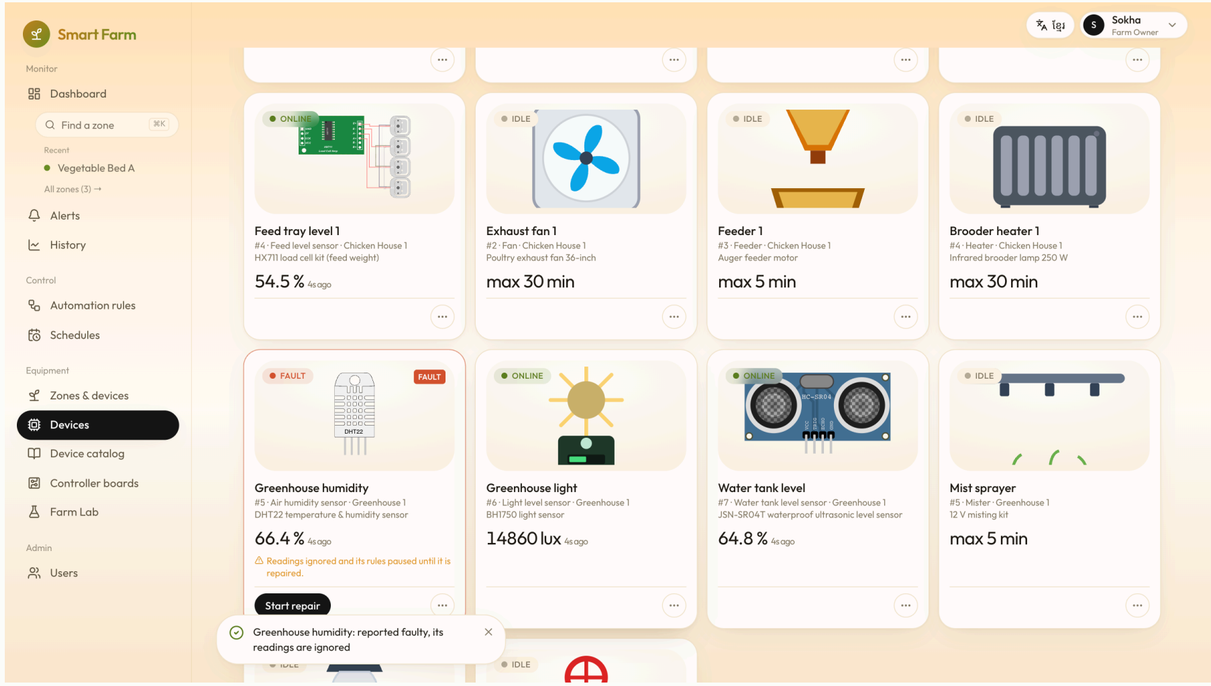
Report fault

Its readings are ignored and its rules wait, so a wrong value cannot start the pump.

2

Repair and test

The technician fixes it; it is back only after a good reading arrives.

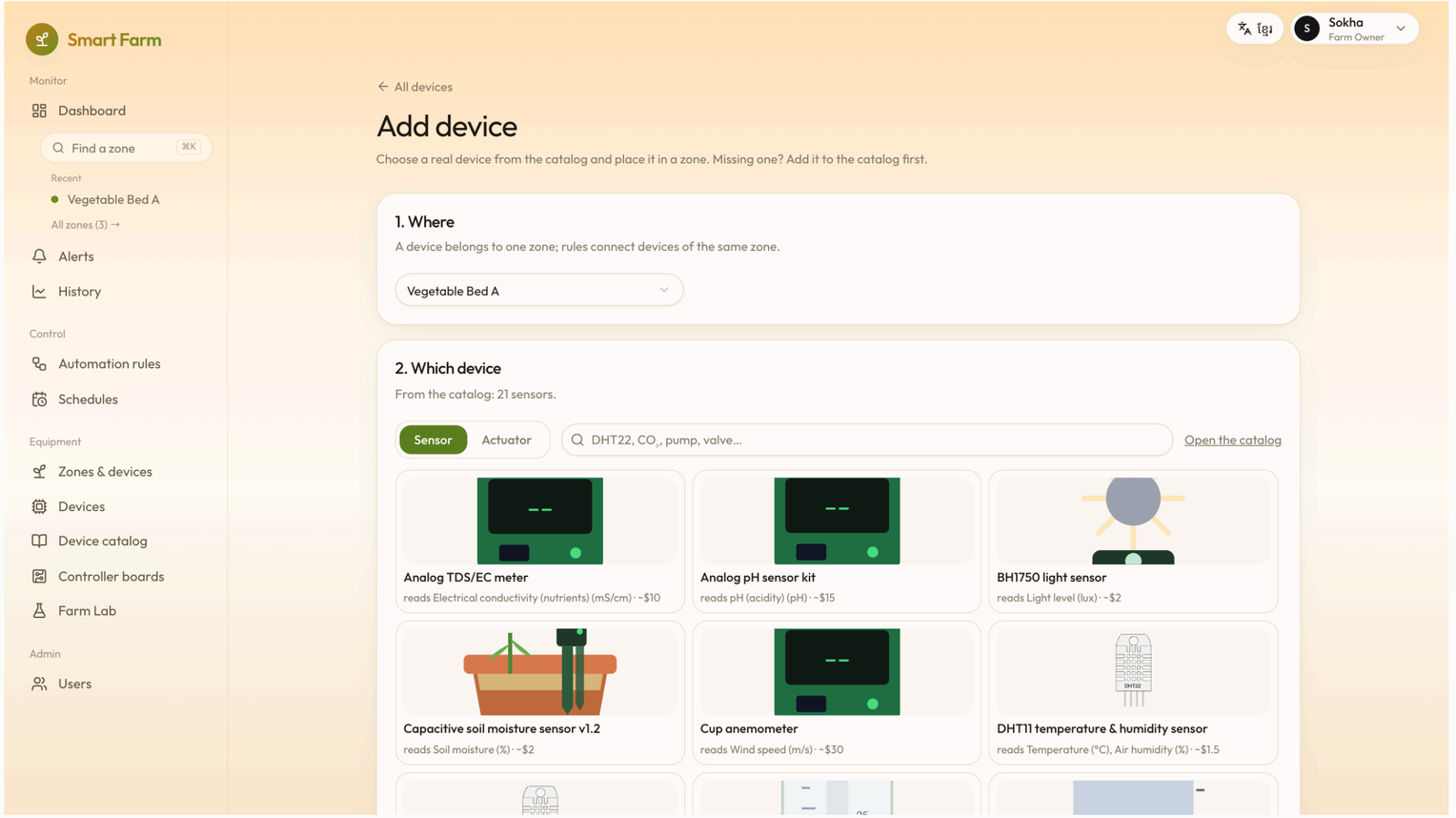


EQUIPMENT

Adding a device takes one page

Pick the zone and the kind of device; each kind says what it is for.

After saving, the app shows how to connect the board and offers to make a rule.

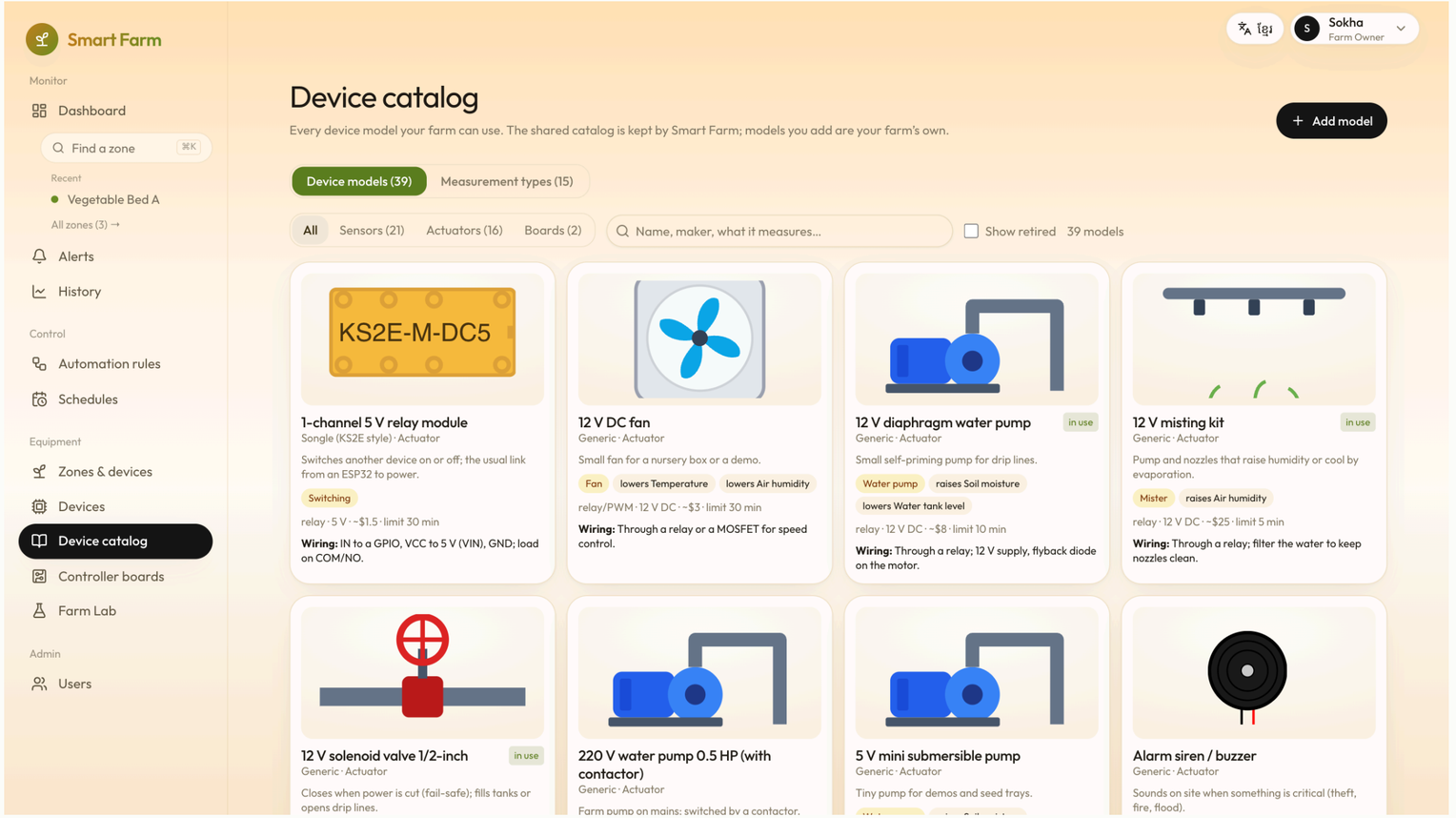


EQUIPMENT

Choose hardware from the catalog

About 40 common sensors and machines, with a picture, what they measure, voltage and price.

A farm can add its own models too.

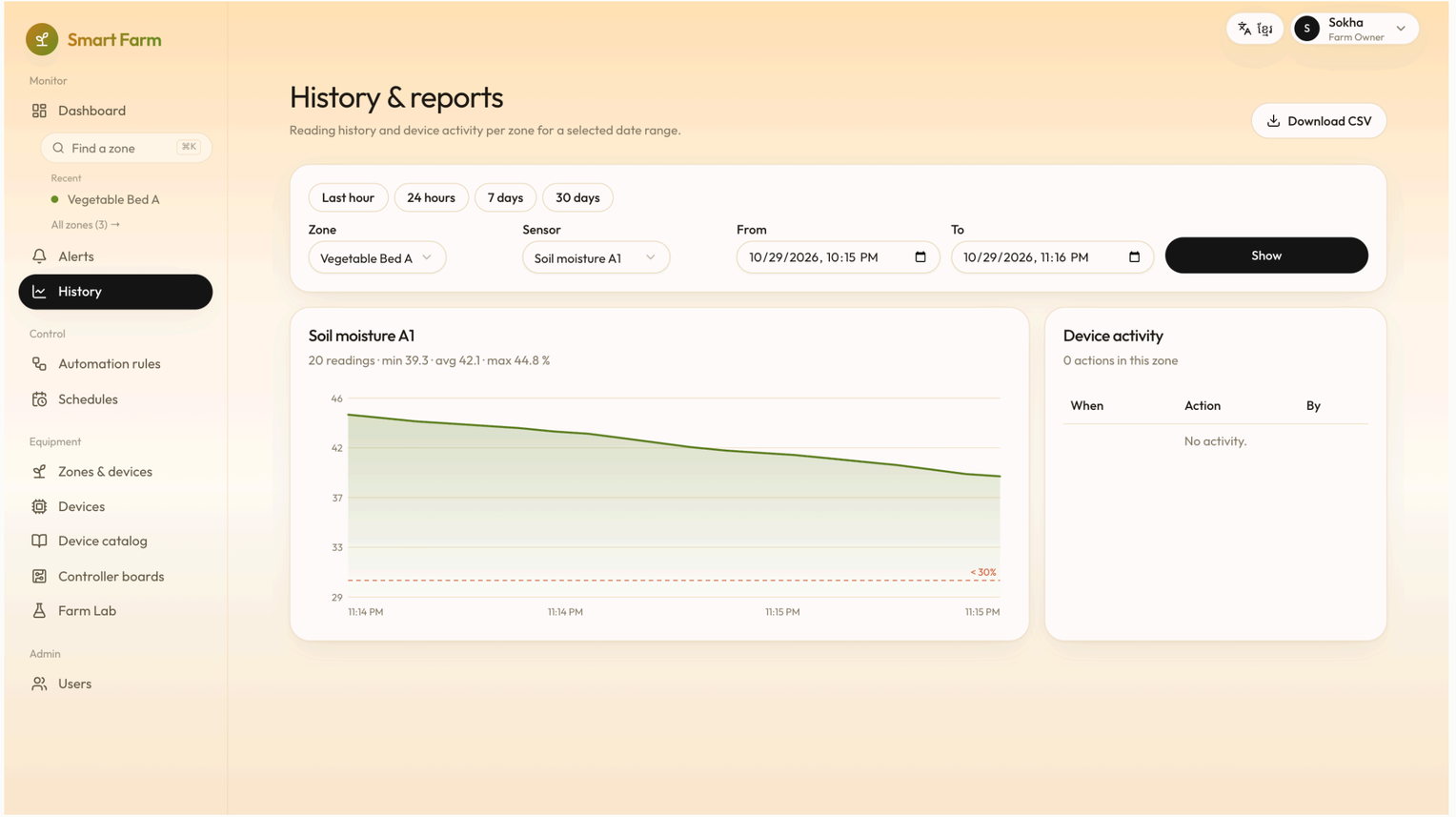


LOOKING BACK

History and reports

Charts of any sensor for a day, a week or your own dates.

Download the readings as a spreadsheet file (CSV).

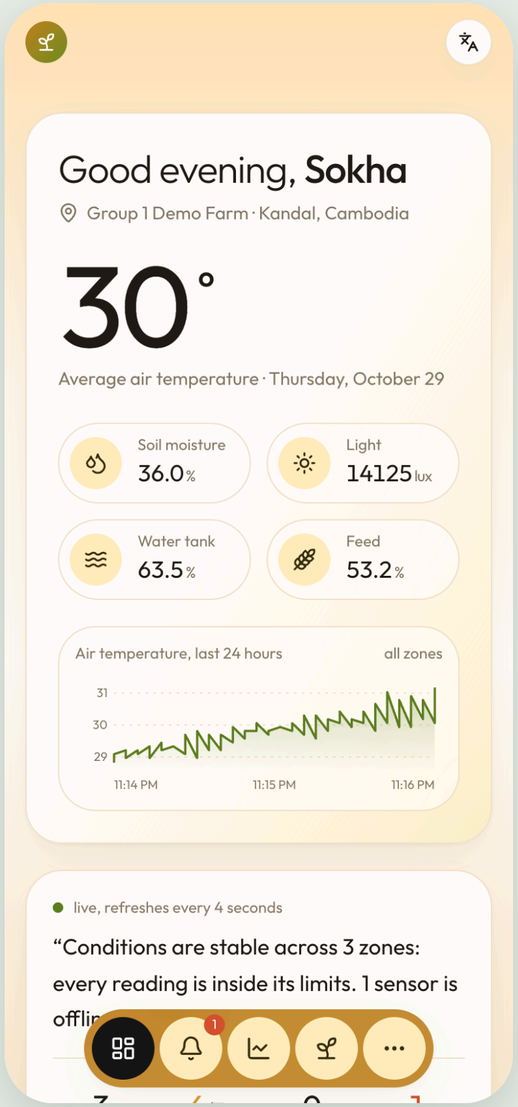


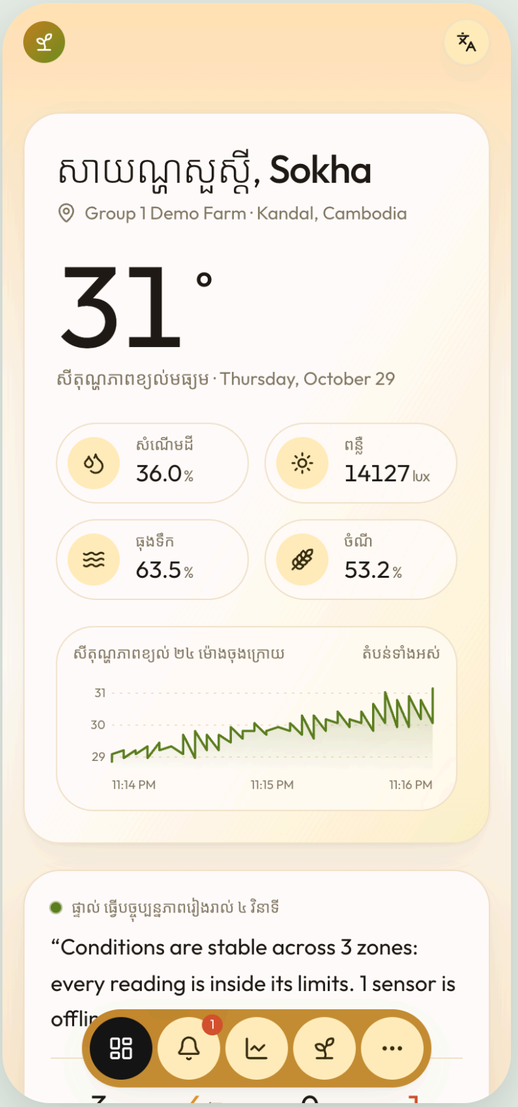
ANYWHERE

On your phone, in Khmer or English

The same app works on a phone, with a tab bar at the bottom.

Switch the language in one tap; the app remembers it.



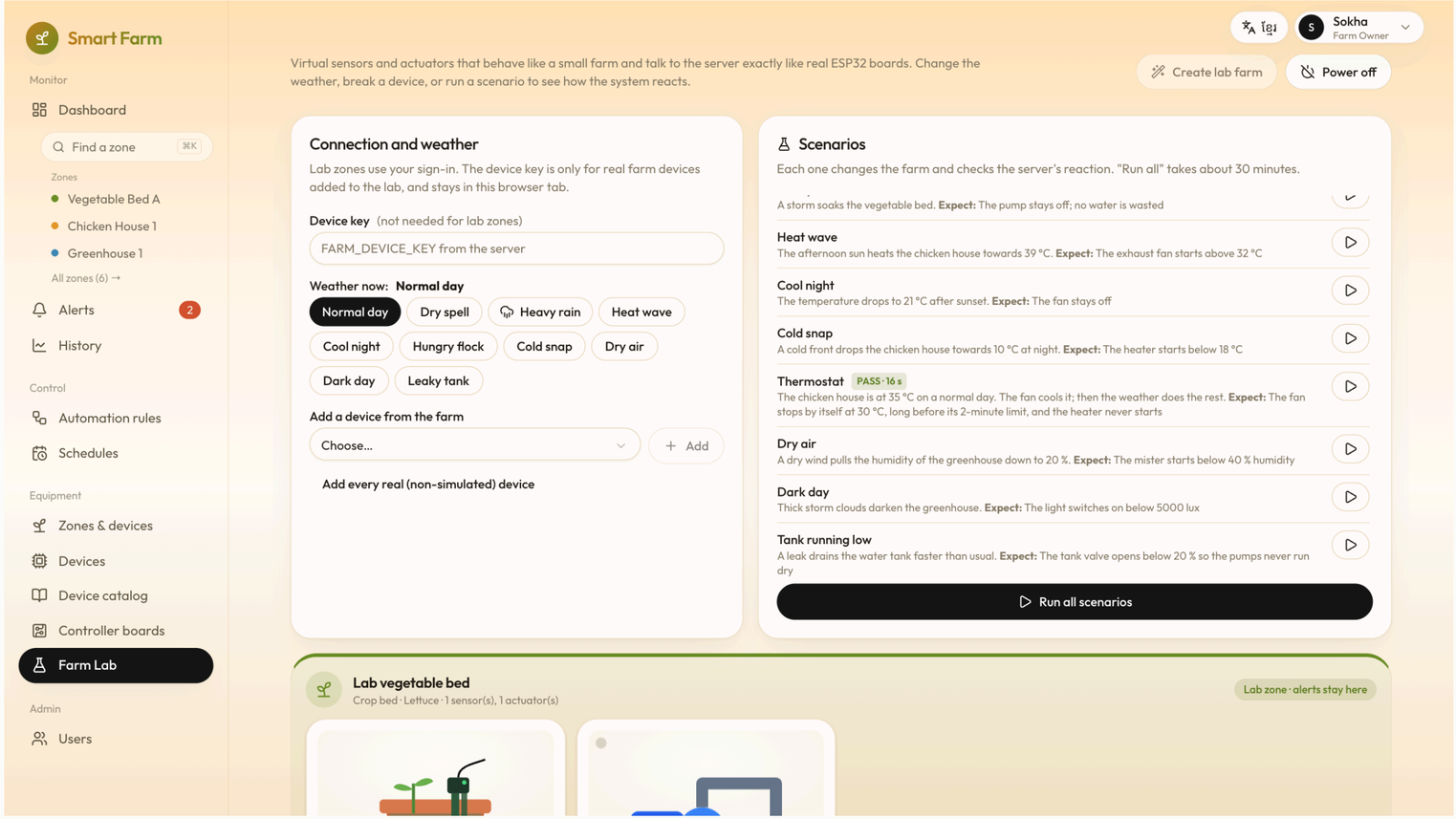


TRY IT FIRST

Farm Lab: practise without any hardware

Virtual sensors and machines behave like a small farm.

Play a dry spell, a heat wave or a power cut and see how the system reacts.

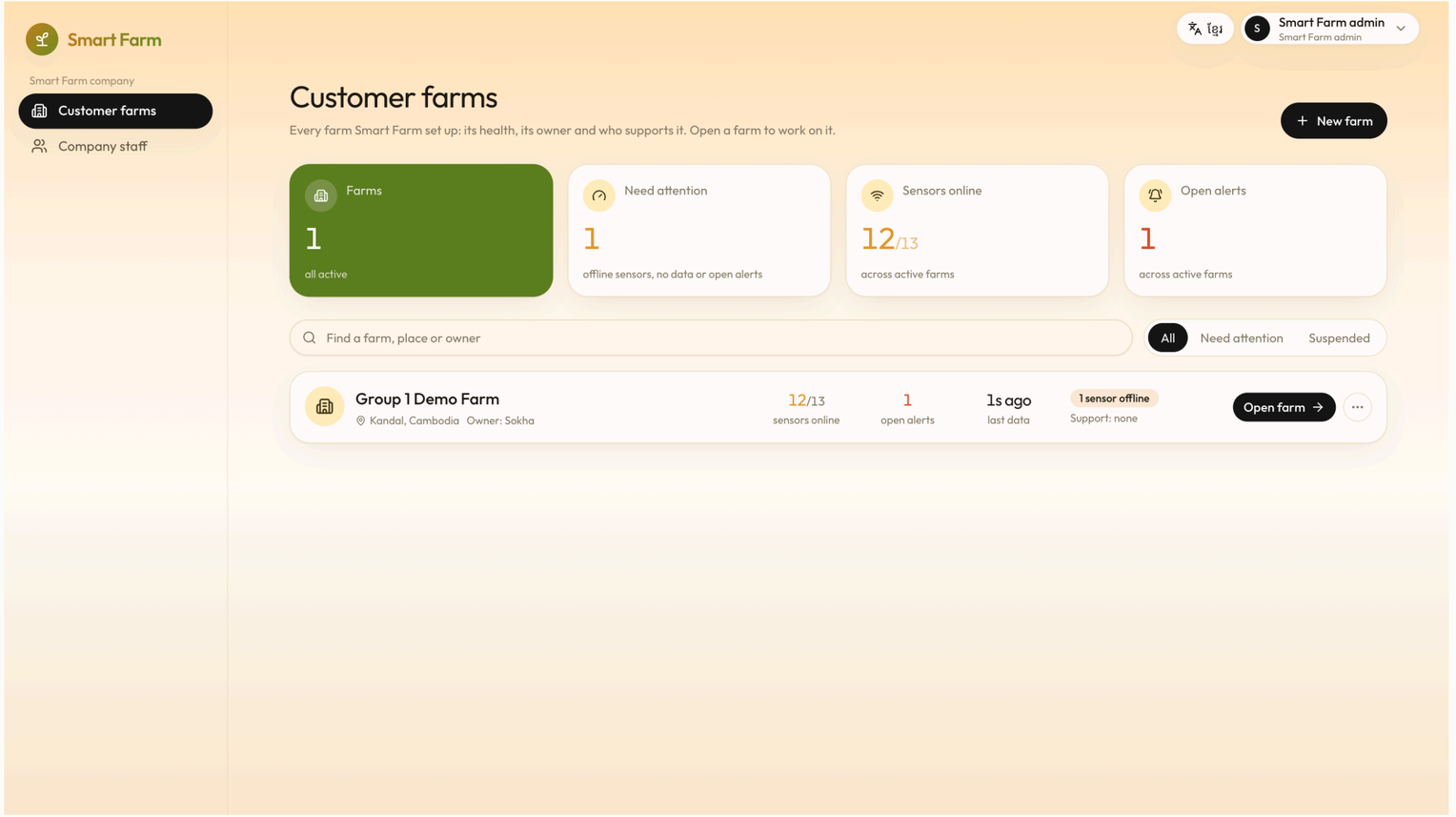


FOR THE COMPANY

Many customer farms, one system

The company sets up each farm and invites its owner with a link.

Each farm only ever sees its own data; support helps only when the owner allows it.



PEACE OF MIND

Safe, private and tested

1

Accounts

Own password for everyone. Five wrong tries lock the name for 15 minutes. Staying signed in survives a server restart.

2

Privacy

Each farm sees only its own zones, devices and alerts. No passwords or keys are kept in plain text.

3

Tested

73 automatic checks, a 34-step walk through every screen, and 17 farm situations like drought and power cuts.

IN SHORT

Less walking, faster reaction, fewer losses

Sensors watch → rules react → you are told → everything is recorded